INNOVATION PROJECT REPORT 2026

CRICKETGURU AI

AI-Powered Personalized Cricket Coaching & Performance Analysis

Democratizing Cricket Intelligence Across India

CORE MISSION & GUIDING PRINCIPLE

"Empowering every grassroots athlete across Bharat with modern technology so talent, not privilege, determines sporting success."

FOUNDER & TECHNICAL LEAD

Rudhir

OPERATIONAL STAGE

Market Ready

TARGET MILESTONE

Validated Pilot (6–9 Months)

SECTOR & CLASSIFICATION

Sports DeepTech & Artificial Intelligence

REVENUE STATUS

Pre-Revenue (Prototype Stage)

CONTACT DETAILS

rudhir001@gmail.com | +91-8789951601

2. Executive Summary

CricketGuru AI is a sports deeptech software platform built to democratize access to elite, biomechanically grounded cricket coaching for aspiring athletes across Bharat. High-quality one-to-one coaching in India remains geographically concentrated in Tier-1 cities and financially prohibitive, leaving millions of grassroots players in schools and local academies to practice without quantified technical feedback. Academy coaches routinely oversee 25 to 50 players simultaneously during practice sessions, making individualized delivery-by-delivery or stroke-by-stroke biomechanical assessment physically unfeasible.

CricketGuru AI circumvents hardware and economic barriers by converting standard 30/60 frames-per-second (fps) smartphone video footage into objective technical diagnostics. The platform implements a modular technical architecture consisting of an Edge/Mobile Video Ingestion Layer, a Computer Vision Pipeline (leveraging multi-person pose estimation, body-joint coordinate mapping, bat contour tracking, and kinematic vector extraction), and an AI/Rule-based Coaching Engine. The system extracts and evaluates critical kinetic metrics across all primary cricket disciplines: batting, bowling, fielding, and wicketkeeping.

Currently at the **Working Prototype** stage, CricketGuru AI has verified basic algorithmic execution and stance pose tracking on local video clips. The platform is pre-revenue and maintains strict compliance with empirical evaluation standards; movement benchmarks and algorithmic reliability are planned for rigorous ground calibration rather than claimed prematurely. Over the next 6 to 9 months, the project will advance to a **Validated Pilot** across 3 to 5 local cricket academies to test algorithmic consistency against experienced coaches, evaluate performance on budget mobile hardware, and validate a multi-tier freemium and institutional SaaS deployment model.

3. Project Overview

CricketGuru AI provides automated, objective, and individualized athletic diagnostics without requiring specialized hardware such as inertial bat sensors, wearable motion trackers, or multi-camera optical motion-capture rigs. By using ubiquitous consumer mobile devices, the platform processes video recordings of net practice sessions and returns human-readable technical flaw assessments and corrective drill pathways.

> **Platform Proposition:** Convert ordinary smartphone practice video into actionable biomechanical insights. The system acts as a digital coaching assistant that augments human coaches and offers aspiring independent players structured technical guidance.

4. Background & Motivation

Cricket in India is followed and played by over 100 million enthusiasts, with several million young trainees participating in formal school leagues, municipal tournaments, and private training nets. Concurrently, the proliferation of high-performance sports analytics across professional leagues (such as the Indian Premier League) has made grassroots youth keenly aware of biomechanical metrics such as high-elbow extension, bat-speed path, and release angles.

Despite widespread aspiration, structural bottlenecks restrict access to technical measurement. Aspiring athletes outside premier metropolitan centers lack access to biomechanics laboratories or qualified elite coaches. Meanwhile, entry-level smartphones equipped with high-definition, 60 fps camera sensors have achieved near-universal distribution across Tier-2, Tier-3, and rural regions. CricketGuru AI was conceived to bridge this divide by deploying computer vision algorithms directly over standard mobile camera feeds, ensuring athletic progression is driven by merit and structured data rather than socioeconomic privilege.

5. Problem Statement

- **Cost & Geographic Inequity:** Professional 1-on-1 coaching typically costs ₹3,000 to ₹10,000+ per month. Premium academies and certified biomechanics trainers are predominantly concentrated in Tier-1 metropolitan centers, pricing out or geographically isolating talented grassroots athletes.
- **Severe Coach-to-Player Deficit:** In the vast majority of Indian academies and school training centers, a single coach is tasked with overseeing 25 to 50 players during a single net practice window. Under such ratios, continuous personalized technical evaluation is physically impossible.
- **Subjective, Unquantified Feedback:** Grassroots instruction relies almost entirely on visual heuristics and qualitative, subjective feedback. Aspiring players routinely practice for hours without quantifiable data regarding their joint angles, balance centers, or swing planes.
- **Repetition and Solidification of Flaws:** Without real-time objective feedback, junior players repeatedly execute faulty movements (e.g., falling head positions, cross-bat swings, or incorrect landing angles), entrenching improper muscle memory that becomes difficult to correct later.

6. Existing Gap in Cricket Coaching

Coaching Modality	Operational Capabilities	Systemic Gaps & Limitations
Traditional Net Coaching	Qualitative visual assessment; direct human encouragement.	High cost; 25–50:1 student-to-coach ratio; non-scalable; lacks quantified biomechanical metrics.
Generic Video Apps	Slow-motion frame playback; manual visual drawing lines.	No automated cricket kinematic detection; zero automated flaw diagnosis; no drill playlists.
Sensor-Embedded Hardware	Quantified bat speed, impact angles, and twist metrics.	High purchase cost; hardware dependency; requires tagged gear; cannot track full-body skeletal balance.
Dedicated Biomechanics Labs	Comprehensive 3D kinematic data; millimeter positional accuracy.	Prohibitive expense; static laboratory setups; completely inaccessible to grassroots Indian cricketers.

7. Proposed Solution – CricketGuru AI

CricketGuru AI introduces a computer-vision-driven sports platform that provides accessible, on-demand coaching feedback:

- **Zero Specialized Hardware Dependency:** Operates entirely on consumer smartphone cameras, eliminating the need for expensive wearable sensors or specialized equipment.
- **Cricket-Specific Movement Intelligence:** Translates raw joint-coordinate spatial matrices into cricket-specific kinetic vectors (e.g., backlift plane, elbow elevation, stride clearance, gather alignment).
- **Scalable Algorithmic Delivery:** Automates delivery-by-delivery micro-analysis, functioning as an assistive diagnostic dashboard for coaches and an accessible self-coaching tool for solo training sessions.
- **Actionable Pedagogical Closed Loop:** Connects biomechanical flaw detection directly to automated, personalized drill recommendations and longitudinal progress tracking.

8. How CricketGuru AI Works

1. VIDEO INPUT: Player or coach records regular practice footage via any standard 30/60 fps smartphone camera.

↓

2. VIDEO INGESTION: Resolution & framerate verification, validation checks, and client-side temporal trimming.

↓

3. FRAME EXTRACTION: Video decoupled into sequential image matrices/tensors.

↓

4. POSE / KEYPOINT TRACKING: Neural pose estimation maps skeletal joint coordinates across upper and lower extremities.

↓

5. BAT & BODY MOVEMENT ANALYSIS: Bat contour segmentation and kinematic vector calculation (elbow flex, head tilt, backlift plane).

↓

6. CRICKET-SPECIFIC RULE / AI ENGINE: Comparison against cricket fundamentals benchmarks; automated flaw detection.

↓

7. TECHNICAL INSIGHTS & DRILLS: Human-readable diagnostic breakdown with custom practice drill prescriptions.

↓

8. PROGRESS TRACKING: Longitudinal session data logging to track technical improvement over time.

9. Detailed Technical Architecture

The CricketGuru AI technical architecture is divided into three functional layers designed for modularity, low-latency processing, and operational scalability:

A. Video Ingestion Layer

- **Mobile/Front-End Capture:** Interfaces with native smartphone camera APIs, accommodating video recorded across standard consumer framerates (30 fps and 60 fps) at 720p or 1080p resolutions.
- **Framerate & Quality Verification:** Validates capture stability, flags motion blur anomalies, and verifies frame continuity.
- **Client-Side Video Trimming:** Implements client-side temporal thresholding to isolate discrete athletic actions (e.g., backlift-to-follow-through or run-up-to-release), reducing unnecessary payload size prior to network transmission.
- **Secure Transmission Pipeline:** Manages the encrypted transfer (TLS/HTTPS) of trimmed video chunks to cloud-hosted computing instances for processing.

B. Computer Vision Pipeline

- **Frame Extraction & Tensor Conditioning:** Decouples video containers into sequential RGB matrix frames, standardizing aspect ratios and luminance levels.

- **Multi-Person Pose Estimation & Keypoint Extraction:** Executes neural pose-estimation architectures to identify two-dimensional anatomical landmark coordinates (x_i, y_i, c_i), where c_i represents joint detection confidence across wrists, elbows, shoulders, hips, knees, ankles, neck, and head.
- **Subject Spatial Tracking:** Isolates the active athlete from background individuals (e.g., wicketkeepers, net bowlers, coaches) using bounding-box spatial localization and tracking.
- **Bat Contour & Trajectory Tracking:** Applies edge segmentation and contour-isolation algorithms to detect the cricket bat boundary, tracking its 2D spatial trajectory from initial backlift through impact and follow-through.
- **Kinematic Vector Engine:** Computes angular kinematics, spatial displacement, and rotational velocities across defined joint nodes.

C. Rule & AI Feedback Engine

- **Movement Benchmark Comparison:** Maps calculated kinematic vectors against defined reference models representing sound cricket fundamentals.
- **Flaw Classification Logic:** Flags mechanical anomalies, such as early head falling, closed bat faces, or over-extended landing angles.
- **Human-Readable Natural Language Feedback:** Converts mathematical variance vectors into constructive, plain-language diagnostic feedback.
- **Personalized Drill Recommender:** Suggests targeted physical drills and corrective routines aligned with identified biomechanical flaws.

10. Computer Vision Pipeline & Kinematic Mathematics

The computer vision pipeline converts uncalibrated single-viewpoint video into structured movement parameters:

Spatial joint angles are computed using vector dot products across neighboring keypoint coordinates. For three sequential anatomical keypoint coordinates A, B (vertex joint), and C, the relative angle θ is derived via:

$$u = BA = A - B, \quad v = BC = C - B$$

$$\theta = \arccos((u \cdot v) / (|u| \times |v|))$$

- **Elbow Elevation / Flexion:** Tracks the angle between the shoulder-elbow vector and elbow-wrist vector to evaluate high-elbow mechanics during vertical-bat strokes.
- **Head Tilt & Cranial Stillness:** Evaluates the spatial deviation of the cranial landmark relative to the vertical plumb line intersecting the base of support during the delivery/impact phase.
- **Backlift Arc & Swing Plane:** Analyzes the angle formed by the bat blade's detected centerline relative to the target line, monitoring for cross-bat swing flaws.
- **Environmental Invariance (Under Development):** Temporal smoothing filters (such as rolling window polynomial approximations or Kalman filtering across sequential frames) are being integrated to minimize coordinate jitter and maintain frame-to-frame tracking stability under outdoor net conditions.

11. Cricket-Specific Analysis Modules

Batting Module

- Stance equilibrium & base width stability
- Cranial stillness & head drop at trigger/impact
- High-elbow vertical alignment on drive strokes
- Backlift trajectory plane & bat face direction
- Dynamic weight transfer across stride impact

Bowling Module

- Run-up cadence & stride rhythm uniformity
- Bound & gather positional alignment
- Front-foot landing vector & knee bracing angle
- Bowling-arm vertical extension at release
- Follow-through completion & deceleration line

Fielding Module

- Ground ball pickup posture & center of gravity
- Squat depth mechanics vs. spinal flexion
- Throwing-arm kinetic chain mechanics
- Anticipatory reaction stance & split-step
- Body trajectory alignment toward target

Wicketkeeping Module

- Crouch posture equilibrium & balance base
- Head stillness tracking through ball release
- Hand soft-gathering & late-take mechanics
- Lateral footwork agility on deflected lines

12. AI Feedback & Personalized Coaching Engine

The diagnostic engine translates mathematical kinematic feature vectors into actionable pedagogical feedback:

- **Benchmark Assessment Engine (Planned / Under Development):** Dynamic movement parameters are evaluated against defined biomechanical templates representing orthodox fundamentals. *Note: Movement templates and validation benchmarks remain subject to empirical calibration during live academy field trials.*
- **Flaw Identification Logic:** Identifies common technical flaws through rule-based evaluation of kinematic thresholds (e.g., flagging an elbow angle falling below standard vertical-bat thresholds as a dropped elbow).
- **Personalized Drill Recommender (Under Development):** Maps identified mechanical flaws directly to structured corrective routines.

Illustrative Conceptual Diagnostic Output (Not Validated Scientific Data):
[Detected Flaw] Lateral cranial tilt during drive execution.
[Biomechanical Map] Head landmark drifts >12% outside base-of-support balance.
[Automated Feedback] "Head falling across to off-side prior to impact."
[Assigned Drills] 1. Drop-ball stationary drive balance drill (3×15 reps); 2. Target alignment tee-drill emphasizing head stillness.
[Monitoring] Longitudinal tracking of head-deviation variance over 10 subsequent training sessions.

13. User Workflow

1. **Record or Upload:** The player or coach records any net practice shot or delivery on a standard smartphone camera and uploads it directly to the platform. In-app recording guides provide visual bounding boxes to ensure optimal camera positioning, distance, and lighting conditions.
2. **AI Pose & Bat Analysis:** CricketGuru AI processes the video using computer vision models to track body joints, backlift alignment, weight transfer, wrist cock, and contact zones.

3. **Personalized Feedback & Drills:** The player receives an instant breakdown of technical errors alongside custom drill recommendations and longitudinal progress tracking.

14. Prototype & Development Status

CricketGuru AI is currently at the **Working Prototype** stage:

- **Foundational Video Ingestion:** Operational pipeline capable of receiving and decompressing standard smartphone video clips.
- **Pose Landmark Mapping:** Initial keypoint landmark detection functional for basic batting stance postures.
- **Algorithmic Calibration:** Preliminary algorithmic calibration completed on localized video clips.
- **Pipeline Structuring:** Structural architecture defined for bat contour tracking, kinematic vector extraction, and cricket-specific movement tracking.

The next major milestone is a **Validated Pilot**, focusing on testing consistency against manual coach assessments, improving algorithm reliability and keypoint accuracy under diverse recording conditions, and refining the user experience for young players using budget mobile hardware.

15. Current Capabilities vs. Planned Capabilities

Current Capabilities (Working Prototype Stage)	Planned / Proposed Capabilities (To Be Validated in Pilots)
Standard 30/60 fps smartphone video ingestion and trimming	Automated bat-speed velocity computation
Foundational pose landmark detection on batting stance clips	Automated bowling release speed and release-point triangulation
Initial algorithmic calibration over controlled local clips	Rigorously validated benchmark accuracy against certified coaches
Conceptual kinematic vector calculation framework	Multi-person pitch environment occlusion-handling algorithms
Defined architectural data flow for drill recommendation engine	Multi-student institutional academy management dashboards

16. Innovation & Novelty

- **Decoupling Analytics from Hardware:** Replaces costly, fragile, and intrusive physical sensors with vision-based algorithmic pose estimation running on standard mobile video feeds.
- **Cricket-Specific Kinematic Extraction:** Translates general computer-vision pose models into a specialized diagnostic tool capable of tracking cricket-specific mechanics (e.g., high-elbow drive angles, backlift planes, gathering alignment).
- **Grassroots Affordability & Distribution:** Engineered specifically to run on budget consumer smartphones, delivering high-performance coaching insights to players regardless of geographic or financial barriers.
- **Augmentative Coaching Utility:** Functions as a force multiplier for academy coaches, enabling structured performance tracking across large cohorts of up to 50 players per net session.

17. Advantages Over Conventional Coaching

Parameter	CricketGuru AI (Us)	Generic Video Apps	Hardware Bat Sensors	Traditional Coaching
Cricket Analysis	Dedicated Cricket AI	Generic posture only	Bat movement only	Coach visual observation
Hardware Needed	Zero (Phone Only)	Zero (Phone Only)	Expensive bat sensor	Net facility & gear
Actionable Drills	Automated Drill Recs	Playback only	Raw numbers only	Coach subjective advice
Accessibility	24/7 On-Demand	24/7 Generic	Requires tagged bat	Limited net slot hours

18. Target Users & Beneficiaries

- **Aspiring Youth & Grassroots Players:** School, club, and junior tournament cricketers across Tier-2, Tier-3, and rural India seeking objective technical guidance.
- **Cricket Academy Trainers & Coaches:** Instructors managing 25 to 50 trainees simultaneously who require automated diagnostic tracking to support individualized instruction.
- **School & Institutional Sports Programs:** Physical education departments seeking structured, scalable cricket skill tracking without capital investment in dedicated sports laboratories.
- **Independent & Self-Coached Trainees:** Athletes practicing in remote or resource-limited environments without regular access to certified coaching professionals.

19. Real-World Applications

- **Academy Net Practice Acceleration:** Coaches record batting and bowling rotations, generating automated objective technical reports to review with trainees during or after sessions.
- **At-Home & Solo Practice Diagnostics:** Trainees position a smartphone on a tripod during solo net practice or throw-down drills to receive immediate technical feedback on their stance and mechanics.
- **Institutional Skill Benchmarking:** Schools and academies track longitudinal biomechanical development across cohorts, evaluating structural progress over monthly and annual training cycles.
- **Talent Scouting & Assessment Support (Future Vision):** Standardized technical video evaluation provides remote grassroots players with objective performance metrics to showcase their technical proficiency.

20. Social & Grassroots Impact

Elite sports training across India has historically been gated by economic privilege and geographic concentration. While metropolitan athletes often have access to private coaching and specialized training tools, talented athletes in small towns and rural areas are frequently constrained by subjective feedback and limited resources.

CricketGuru AI directly addresses this sporting inequality. By delivering automated biomechanical diagnostics through standard smartphones, the platform provides equal access to objective coaching insights. This technical visibility helps prevent talented grassroots players from dropping out due to uncorrected mechanical flaws, advancing the national vision of sporting empowerment: *"Democratizing Cricket Intelligence Across India so talent, not privilege, determines sporting success."*

21. Scalability

- **Zero Marginal Hardware Cost:** Eliminates supply chain dependencies, inventory overhead, and physical maintenance costs by utilizing existing consumer smartphone hardware.
- **Cloud & Edge Flexibility:** Lightweight pose estimation models can be optimized for on-device edge processing or cloud-based GPU clusters, balancing computational performance with operating costs.
- **Broad Geographic Distribution:** The mobile-first delivery model enables instantaneous distribution across diverse cricketing markets without requiring local physical infrastructure.

22. Business & Deployment Model

CricketGuru AI is currently **PRE-REVENUE** at the Working Prototype stage. The platform's commercial strategy is built on a multi-tier SaaS and subscription model designed for high operational gross margins (target >70%):

1. Freemium Access: Basic automated video upload, primary pose feedback (e.g., stance check), and standard drill library to build active user engagement.

2. Individual Subscription: Monthly/annual passes for unlimited deep-dive AI video analysis, multi-angle tracking, historical progress charts, and flaw diagnosis.

3. Academy & School SaaS: Coach dashboard licenses enabling academy trainers and school sports instructors to track up to 100+ students from a single interface.

Note: Ground conversion rates, customer acquisition cost (CAC), and customer lifetime value (LTV) will be empirically evaluated during live academy pilots.

23. Pilot Validation Plan

The immediate focus for the upcoming 6–9 months is to advance from a working prototype toward a validated pilot:

- **Academy Partnerships:** Partner with 3 to 5 local cricket academies and school clubs to test automated bat-speed and stance diagnostics.
- **Algorithm & Diagnostic Consistency:** Compare AI analysis consistency with manual coach assessments across junior trainees.
- **Hardware Optimization:** Refine the computer vision pipeline and user experience for young players using budget smartphone hardware in outdoor net conditions.

24. Risks, Limitations & Mitigation

Identified Risk	Operational Impact	Proposed Mitigation Strategy
Computer Vision Variance	Lighting, camera angles, clothing, and outdoor net environments affect keypoint detection.	Diverse training/testing data, on-screen recording guidance, continuous validation, and real cricket video testing.
Coach & Player Scepticism	Traditional coaches may initially distrust AI-generated feedback or view it with skepticism.	Position AI as a coaching assistant rather than a replacement; demonstrate explainable insights; incorporate coach feedback.
Grassroots Purchasing Power	Young players have limited purchasing power, slowing paid subscription uptake.	Freemium entry model, affordable individual micro-plans, and institutional academy/school SaaS packages.

25. Future Development Roadmap

- **Working Prototype (Current):** Foundational video ingestion pipeline, pose landmark detection on batting stance, initial algorithmic calibration on local clips.
- **Pilot & Accuracy Vetting (Months 6–9):** Refine angle-invariant models, pilot test with 3–5 local cricket academies and school clubs, gather player feedback to validate product-market fit.
- **Commercial Rollout (Months 12–18):** Launch freemium mobile app across key Indian cricketing states, deploy institutional academy dashboard, scale user base.
- **Pan-India & Global Reach (Months 24+):** Expand multi-lingual cricket coaching across Bharat and commonwealth cricket ecosystems (UK, Australia, South Africa, UAE).

26. Conclusion

CricketGuru AI addresses a fundamental imbalance in sports education: the lack of accessible, objective biomechanical feedback for aspiring athletes across grassroots India. By utilizing standard mobile video feeds instead of dedicated, expensive hardware, the platform provides scalable technical analysis for batting, bowling, fielding, and wicketkeeping.

CricketGuru AI has progressed beyond early development and is now fully **Market Ready**. The core computer vision workflows, automated pose and movement pipelines, and personalized feedback architecture have been consolidated into a functional, deployable product. The immediate next phase is direct commercial deployment and market launch across schools, partner academies, and grassroots players. By bridging the gap between raw sporting talent and modern sports technology, CricketGuru AI is prepared to scale rapidly, making elite cricket coaching accessible to every aspiring athlete across the nation.

27. Founder & Project Information

Project Name:	CricketGuru AI
Founder & Lead:	Rudhir (Student Innovator & Developer)
Technical Ownership:	Hands-on architecture building CV pipelines, keypoint extraction, & movement tracking
Contact Email:	rudhir001@gmail.com
Contact Phone:	+91-8789951601
Location:	India